

Design and Analysis of a Power-Efficient Leg Mechanism for Agile Small Quadruped Robots

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Abstract

Quadruped robots are becoming increasingly important due to their versatility and adaptability to challenging environments. The leg design of a quadruped robot is of paramount importance, as it directly impacts its performance, stability, and adaptability to various terrains. Existing designs have utilized hydraulic actuators, brushless motors, or servo motors for the joint actuators. While brushless motors and hydraulic actuators provide lots of torque, they often consume lots of power and are costly. For small quadruped robots with limited space and power, usage of such actuators would be inefficient, especially for applications that might require longer battery life such as search and rescue missions. Furthermore, servo motors while maintaining low energy cost and a small form factor, often lack the torque and speed needed to perform rapid maneuvers such as jumping. We propose a design for a quadruped robot leg that uses servo motors but is still able to maintain enough torque, speed, and range of motion to enable agile maneuvers. We validate the design using a series of tests that ensure the design reaches the required performance.

Keywords quadruped robot, structural design, actuators, legged robot, mechanical design

Paper Type Technical paper

1. Introduction

Quadruped robots have potential applications in various fields, including search and rescue, exploration, and logistics. These robots, inspired by biological quadruped locomotion, offer unique advantages such as superior terrain adaptability and obstacle avoidance capabilities Fan et al. (2024), Chai et al. (2022). They are becoming increasingly important due to their versatility and ability to operate in challenging environments where wheeled robots struggle.

The leg design of a quadruped robot is of paramount importance, as it directly impacts its performance, stability, and adaptability to various terrains. Existing leg designs for quadruped robots have made significant strides, but there is still considerable room for improvement in terms of energy

efficiency and dynamic performance. Many existing designs depend on intricate mechanisms or powerful actuators, such as hydraulic systems or brushless motors, leading to increased weight, higher costs, and greater maintenance demands Raibert et al. (2008), Semini et al. (2015), Kau et al. (2019). For small-scale quadruped robots with limited space and power budgets, the use of powerful but energy-intensive actuators becomes highly inefficient, particularly for applications requiring longer operational times, such as search and rescue missions.

Conversely, while servo motors are widely used due to their low energy cost and compact form factor, they often lack the torque and speed required to perform rapid, agile maneuvers like jumping. This presents a critical design challenge: achieving a balance between energy efficiency and agility on a limited power budget remains a difficult consideration.

This paper proposes a novel design for a quadruped robot leg that addresses this very challenge. We present a design that utilizes servo motors but is engineered to maintain sufficient torque, speed, and range of motion to enable agile maneuvers. We validate our design through a series of rigorous tests that

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demonstrate its performance, ensuring it reaches the required metrics for dynamic locomotion and agility.

2. Literature Review

One of the earliest implementations of a design using hydraulic actuators is BigDog Raibert et al. (2008) by Boston Dynamics. BigDog uses an internal combustion engine as a power supply and has four hydraulic actuators that power the joints for each leg. While BigDog's speed can reach up to three m/s, it weighs about 109 kg. This further confirms that hydraulic actuators would be inapplicable for small robots that weigh less than 10 kg.

Hydraulic actuators are actuators that use hydraulic oil to generate mechanical motion such as linear, or rotary motion. They are widely used in various industries due to their high power density. However hydraulic actuators are often heavy and require tanks of fluid to allow the actuators to extend and retract. Hydraulic actuators are also relatively slow compared to brushless motors. These limitations limit hydraulic actuators to larger quadruped robots as smaller quadruped robots can not integrate a hydraulic system without sacrificing space and weight.

HyQ2Max and HyQ2CENTAUR Semini et al. (2015, 2017) are two quadruped robot designs based on hydraulic actuators. These designs are meant to improve a previous design (HyQ) by the same authors Semini et al. (2011). The designs are based on linear hydraulic actuators for hip and knee extension joints while the whole leg is rotated using a brushless motor. To support the hydraulic system required for this actuation strategy, the robot's weight reached 91 kg. The range of motion for this robot is also limited to 120° for each joint. The authors also proposed a smaller design called MiniHyQ Khan et al. (2015). MiniHyQ still weighs around 35 kg and still uses a hydraulic system. Even though it is supposed to be the smallest version of the HyQ series of quadruped robots, it still stands at 770 mm tall.

Recently, brushless motors have been more widely used for quadruped robots. For example, the Stanford Doggo (Kau et al. (2019) uses two brushless motors to control each leg. The motors use belts to transfer the force to the necessary links. The motors are geared down (3:1) to increase the torque. The Stanford Doggo achieved an impressive vertical height jump of 1.14 m. However, the power cost for running this system is high as it has a max continuous power of 840 W. Assuming a 3-cell battery is used, a capacity of 2200 mAh would last only 5 minutes under this continuous power draw. Additionally, each leg requires its own motor driver board (Odrive) which increases the cost and the space required to mount all these boards.

An example of a quadruped robot using brushless motors for joint actuation is MIT's Cheetah series Bosworth et al. (2015), Katz et al. (2019), Bledt et al. (2018), Park et al. (2017). All variations of this robot are highly agile; however, this agility comes at the expense of power-hungry actuators. Even the smallest variation i.e., the Super Mini Cheetah,

requires two 12-cell (44.4 V) batteries for operation, contributing to significant energy consumption that limits the robot's operational duration. Additionally, these actuators are expensive and require specialized controllers to manage motor behavior. The same is true for commercially available quadruped robots such as Anymal Hutter et al. (2016), Boston Dynamics' Spot Bouman et al. (2020), and Unitree's Go2 Robotics (n.d.).

Multiple quadruped robots Saputra et al. (2021), Fredriksson (2021), Shi et al. (2021), Chen et al. (2021, 2020), Rahman et al. (2023), Alex & Nathan (2023) have been developed using servo motors as the joint actuator. However, the motion range for the joints has been limited, and the servo motors' torque is barely enough to withstand the weight of the robot. For example, Fredriksson (2021) used Dynamixel AX-12a servo motors which have 1.5 Nm of torque. The design of the leg also limits the rotation of the motor by using a linkage mechanism to link the lower leg link thus preventing the lower link from reversing to the opposite direction which is useful especially for self-righting and fall recovery. This linkage design can be seen in many similarly sized robots Alex & Nathan (2023), PuppyPi Hiwonder Quadruped Robot with AI Vision Powered by Raspberry P (2024).

To summarize, the existing quadruped robots either maintain the range of motion and agility at the cost of high power consumption as in the case of Stanford Doggo or MIT Cheetah or utilize servo motors which do not consume much power, but have a limited range of motion and agility.

We propose a leg design that utilizes servo motors for joint actuation, thus maintaining low energy costs while still providing enough torque and range of motion to allow agile maneuvers that are otherwise not possible for existing designs using servo motors. The leg design was validated using a series of experiments.

3. Methodology

3.1 Kinematics

To control the quadruped robot leg, the inverse kinematics need to be deduced. The inverse kinematics are based on geometrical relations between the links. The Equations ?? - ?? detail the calculations for the inverse kinematics for the hip rotation joint (Hip_r), hip extension joint (Hip_{ex}), and knee extension joint (Kn_{ex}). *offset* refers to the distance of the link between the motor and the leg for the hip rotation joint. X and Y refer to the target leg position on the ground where X is parallel to the leg's walking direction and Y is perpendicular to that direction. Z refers to the required leg height. len_{I1} refers to the length of the upper link for the leg relative to the knee joint while len_{I2} refers to the length of the lower link relative to the knee joint.

3.2 Dynamics

To design a quadruped robot leg, the number of degrees of freedom was selected to maximize the range of motion. The leg is designed to have three degrees of freedom. Two degrees of freedom for extension and one degree of freedom for rotation.

The motor choice is also very important for the design of the leg. The chosen motor needs to have sufficient torque, speed and range of motion. The design should then try to allow the motor to exercise the full range of motion.

The motor selection process requires the torque to be calculated following the leg dimensions. Two methods are used to calculate the required torque: Newton's equations, and the Newton-Euler method. For Newton's equation, the weight of a single leg can be calculated by dividing the overall weight of the robot by four. This weight can then be multiplied by gravity as shown in Equation ?? . The calculated force is then used to calculate the torque requirements for both the hip and knee motors by multiplying the length of the links as shown in Equations ?? and ??. Assuming the max weight for the proposed robot is 8 kg, the force applied on one leg is 2×9.81 . The torque needed for the hip motor would thus be 4.80 Nm and the torque required for the knee motor would be 2.25 Nm.

The Newton-Euler method [Khalil \(2011\)](#) is an iterative method that involves a forward pass and a backward pass. The forward pass calculates the joint velocities and accelerations. These velocities and accelerations are then used by the backward pass to calculate the force and torque required to move the joints. Equations ??, ??, and ?? are considered the forward pass equations where ω_i represents the velocity of the i th joint and α_i represents the acceleration of the i th joint. R_i^{i+1} is the rotation matrix from the i th joint to the next joint ($i+1$). $\dot{\theta}_{i+1}$ represents the initial velocity that the next joint has.

The inertial force (F_i) and moment (N_i) are then calculated in Equations ??, ??, and ??. I_{ci} is the inertia matrix of the link which can be calculated using the physical properties of the design. m_i is the mass of the link, and a_{ci}^i is the acceleration of the center of the link. P_{ci}^i represents the translation matrix from the centre of the link to its end. The backward pass can then be calculated. The forces and torque for each joint are calculated using Equations ?? and ?? where f_i is the force on each joint and n_i is the torque for each joint. The torque calculated using this method is 5.50 Nm and 2.20 Nm for the hip and knee joint, respectively.

3.3 Design Specifications

Based on the mentioned torque and range of motion requirements, the motor and design components were chosen as shown in Table I. The chosen motor is the 9imod servo motor (55 kg/cm 9imod Servo Motor).

It is usually preferred to keep the motors as close as possible to the quadruped robot's body as this minimizes the torque needed to move the leg. If the motors are mounted further down near the knee joint, then this would put extra load on the hip joints. Therefore a transmission is necessary to transfer torque to the lower link. To avoid using a linkage mechanism for this transmission, a belt transmission is used instead. Slots shown in Fig. 1 in the upper link allow the belt to be tensioned.

The finalized design shown in Fig. 2 allows the knee and hip extension joints to have 270° of rotation allowing them

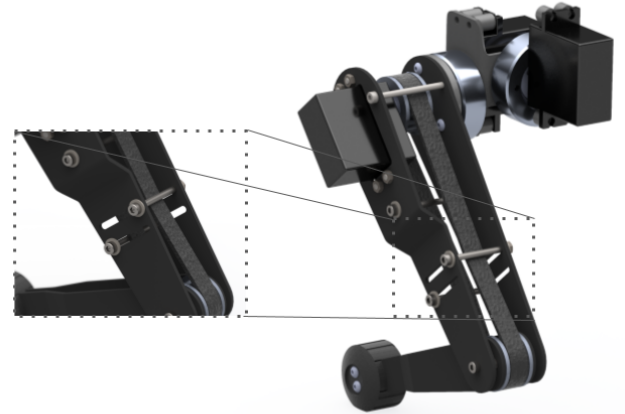


Figure 1: Tension slots for the proposed leg design

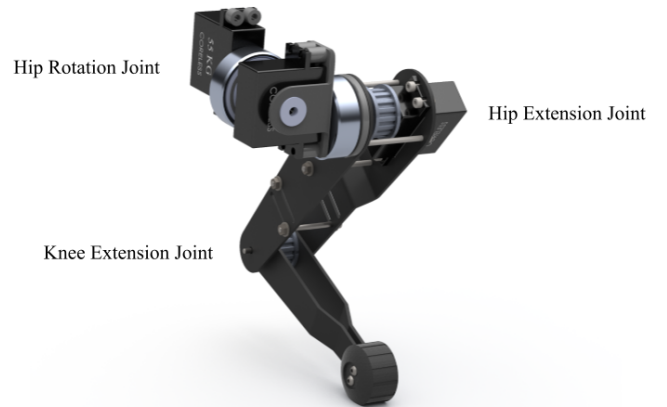
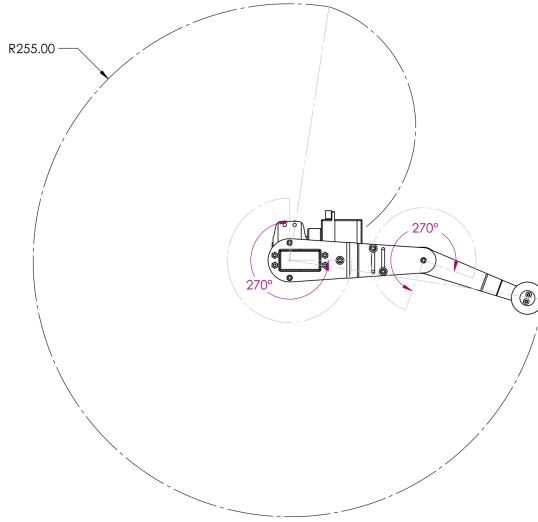
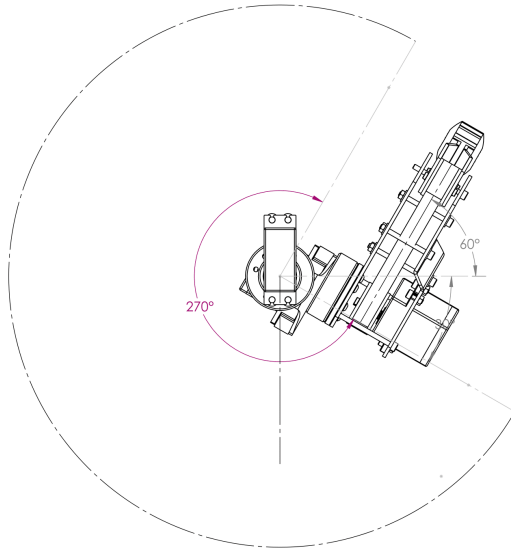


Figure 2: Finalized leg design isometric view

to bend the knee joint to the opposite direction and still be functional. The hip rotation joint is also capable of 270° rotation allowing for very flexible movements as demonstrated in the working envelope shown in Fig. ?? and ??. This range of motion is much larger than existing servo motor designs that rely on linkages to transfer power such as Monash's Dingo [Alex & Nathan \(2023\)](#) robot shown in Fig. ?? and ??. This design allows fall recovery to be possible if the robot unexpectedly loses balance. The design was fabricated using aluminium metal except the edge of the foot which was 3D printed using the TPU material. TPU is a flexible material thus it acts as a passive suspension mechanism for the leg. Some designs [Sun et al. \(2023\)](#) rely on soft materials for the whole leg however this is only applicable to extremely small robots.



(a) proposed leg design side view.



(b) proposed leg design front view.

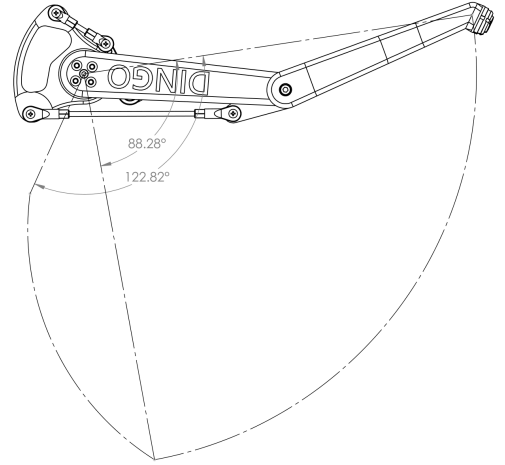
Figure 3: Working envelope for the designed leg for side view(a) and front view(b).

3.4 Experiment Design

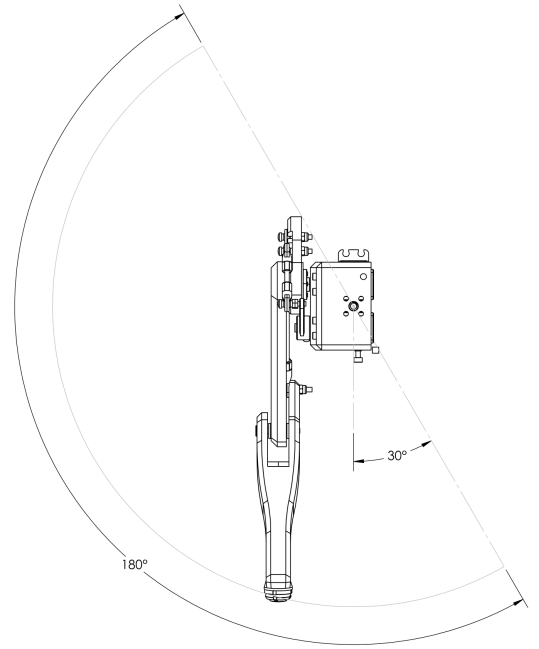
The fabricated leg is connected to a 3D printed slider as shown in Fig. 5. This setup allows the leg to move vertically while restricting horizontal movement. This setup allows for the following experiments to be conducted:

3.4.1 Torque Experiment

In order to test the leg's suitability for agile movement, several tests have been conducted. First, the leg's max weight load was measured by continuously adding weight to the test rig shown in Fig. 5. The leg attempted to change the extension



(a) dingo robot leg design side-view working envelope.



(b) Dingo robot leg design front-view working envelope.

Figure 4: Working envelope for Monash's Dingo Alex & Nathan (2023) robot leg.

height from 150 mm to 100 mm throughout testing. Data from the Inertial Measurement Unit (IMU) was used to determine when the leg's performance has been severely impacted by the load. The IMU sensor is placed in two different configurations. The experiment is carried out with the IMU sensor placed at the hip extension joint and then the experiment is repeated with the sensor at the knee extension joint.



Figure 5: Experiment setup for torque and speed evaluation.

3.4.2 Speed Experiment

The second experiment involves testing the maximum speed at which the leg can be sent commands while maintaining a recognizable pattern. The leg was fed the same loop of commands but with decreasing time between said commands. Using the IMU sensor attached to the leg, the pattern of motion can be observed until the leg fails to follow the commanded pattern. For both experiments, the IMU sensor was placed at two different spots (both extension joints) to eliminate any biases that might occur due to the positioning of the sensor. The IMU sensor also had two different orientations.

Table I: Design Specifications

Motor	9imod Servo Motor
Motor Stall Torque	5.8 Nm
Motor Max RPM	125 RPM
Motor Angle Range	270°
Motor Voltage	8.4 V
Motor Max Current	5 A
Lower Link Transmission	Timing Belt
Mechanical Reduction	1.25:1
Upper Link Length	0.130 m
Lower Link Length	0.115 m
Cost	\$23.23

3.4.3 Power Consumption

The third experiment consists of measuring the voltage, current, and power while the leg is performing repetitive height changes from 0.13 m to 0.15 m. The duration taken to reduce the 2S 2200 mAh battery from 8.4 V to 7.4 V under a load of 500 g was measured to estimate the average power draw. The maximum and minimum power draw were also measured by varying the load on the robot leg. The measurements were taken using a watt meter attached between the battery and the motors.

4. Results

The proposed leg design was able to withstand a maximum load of 2.2 kg and a minimum step time of 50 ms. These values were observed to be the limit as the pattern is distorted at these values. Meaning that the motors are not able to keep up with the load or speed. Both positions for the IMU sensor generated similar results with the difference being the axis of rotation.

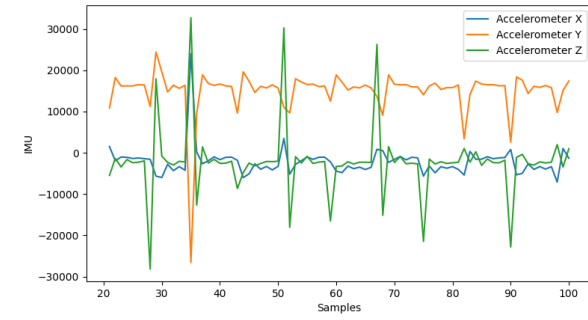
For the torque experiment, the maximum load is 2.2 kg without including the leg's weight which is about 0.5 kg. The graphs in Fig. 6a and 6b show that the pattern of motion was maintained up until it flat lines at 2.2 kg. This means that the whole quadruped robot can weigh up to 8 kg and still be functional. This torque allows the robot to carry additional supplies that might otherwise impede the motion of other robots with servo actuators.

In comparison to the brushless motor used for Stanford Doggo which has a peak torque of 4.80 geared down to 14.4 Nm, the servo motor used for the proposed leg has a maximum torque of 5.30 Nm. The Monash Dingo robot however uses servo motors and has a peak torque of 3.50 Nm. The proposed leg design has higher torque than similar legs using servo motors but still lags behind the approaches that utilize brushless motors.

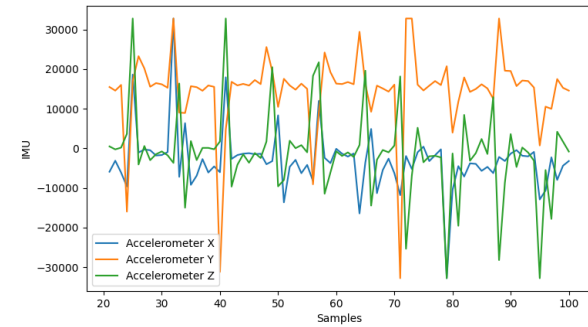
For the speed experiment, it is clear that the pattern becomes irregular at 50 ms and below as shown in the graphs in Fig. 7a and 7b. The speed can be seen following the pattern for the leg movement up until the cycle period of 100 ms where the irregularity starts to appear. Speed is an important aspect of agile movements as sufficient speed is required for acrobatic maneuvers such as jumping. The leg demonstrated the ability to jump up to 0.05 m.

For the power consumption experiment, the proposed leg was able to drain the 2S 2200 mAh battery in 1 hour and 16 minutes. While standing still, the leg draws an average of 0.35 Amperes under a load of 500 g and 1.10 Amperes under a load of 2 kg. In comparison to the Stanford Doggo or MIT Cheetah, which would last approximately 7.40 minutes and 19.8 minutes respectively.

In order to compare the leg design with other quadruped robots in terms of power consumption, the power-to-weight ratio was calculated. The power-to-weight ratio of the proposed leg is 0.021 W/g and a maximum of 42 Watts can be drawn during the movement of the leg under the load of 2



(a)



(b)

Figure 6: Torque results for 2.2kg(a) and 0.5kg(b).

kg. Table II shows a power consumption comparison between similarly sized robots where the proposed leg design has the least power-to-weight ratio making it the most power-efficient design.

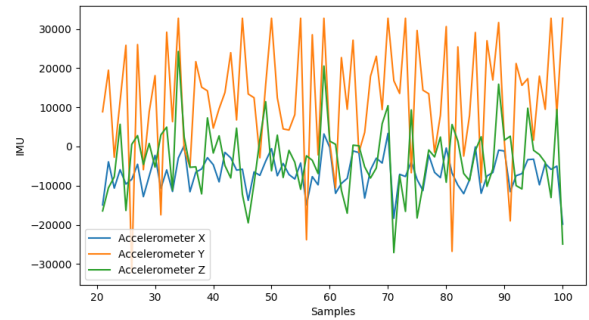
In terms of size, the proposed design is of similar size and weight to the Super Mini Cheetah while requiring much less power consumption. The small size allows for search and rescue missions in tight spaces where the robot can crawl. Table III shows the height and weight of the proposed design in comparison to other quadruped robots.

Table II: Power Consumption Comparison

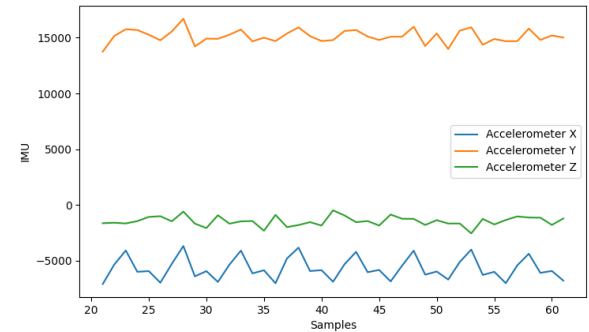
Design	Power to Weight Ratio (W/g)	Max Power Consumed (W)
Proposed Leg	0.021	42
Stanford Doggo	0.698	840
Super Mini Cheetah	0.145	328

5. Discussion

The experimental results validate the proposed leg design's ability to achieve a crucial balance between dynamic performance and energy efficiency using servo motors. The findings demonstrate that while the design may not match the raw



(a)



(b)

Figure 7: Speed results for 50ms(a) and 500ms(b).**Table III:** Size Comparison

Design	Height(cm)	Maximum Weight of Robot(kg)
Proposed Leg	15	8
Stanford Doggo	16	4.8
Super Mini Cheetah	15	9
MIT Cheetah 3	20	45
Anymal	25	30
MiniHyQ	77	35

torque and speed of systems utilizing brushless motors, it surpasses similar servo-driven designs in key performance metrics, all while maintaining superior power efficiency.

The torque experiment confirmed that the leg design is capable of supporting a significant load, with a maximum capacity of 2.2 kg per leg. This is particularly noteworthy when compared to the Monash Dingo robot, which also uses servo motors but has a lower peak torque. The ability of the full robot to handle a total weight of up to 8 kg enables it to carry additional equipment and payloads, expanding its potential applications in logistics and search and rescue missions. While the peak torque of 5.30 Nm does not reach

the 14.4 Nm of the Stanford Doggo's brushless motor system, the trade-off is a substantial gain in energy efficiency, a core objective of this design.

The speed experiments indicate that the leg's performance is optimized for a cycle time of 100 ms and above. Below this threshold, the motors are unable to maintain the desired motion pattern, suggesting a limit to the system's dynamic agility. However, the leg's successful jump of 0.05 m demonstrates that the design possesses the necessary speed and range of motion for agile maneuvers, directly addressing the challenge of enabling such movements with servo actuators.

The most significant finding is the proposed design's power consumption. With a power-to-weight ratio of 0.021 W/g, it is demonstrably more energy-efficient than all comparable robots studied, including the Stanford Doggo and Super Mini Cheetah. This superior efficiency, combined with its compact size (similar to the Super Mini Cheetah), makes the proposed quadruped robot a compelling solution for long-duration missions in confined spaces, such as those encountered in search and rescue. The battery life of 1 hour and 16 minutes further highlights its operational longevity compared to more power-hungry designs.

In summary, the results confirm that this leg design successfully addresses the trade-off between power consumption and agility. It provides enough torque and speed for dynamic locomotion and maneuvers like jumping, while being significantly more power-efficient than other systems. The design proves that it is possible to create an agile and capable quadruped robot for demanding applications without relying on high-power, high-cost actuators.

6. Conclusion

The proposed quadruped leg design can meet the requirements for agile movement while maintaining low power consumption with a maximum required power of 42 W. The proposed design also has achieved a larger range of motion compared to other designs using servo motors by utilizing belts for power transmission and servo motors with 270° of motion.

The power efficiency is 87% less than Super Mini which allows for quadruped robots that can cover large distances without needing to recharge the batteries. The leg design can be used to build search and rescue quadruped robots that are small enough to crawl into spaces and are power efficient enough to maintain a reasonable operation time. The range of motion of the leg allows for recovering from falls or crawling into small spaces. The cost for the components used in this design is also much lower than designs using brushless or hydraulic actuators. Future works may use the designed leg to build a quadruped robot to test the agility and robustness of the robot system.

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